MCMC Diagnostic Appendix

Bayesian AR(p) Engine

2026-04-06

Executive Summary

This appendix provides the diagnostic verification for the Bayesian AR(p) model fit. We assess chain convergence, mixing efficiency, and parameter recovery against the ground truth.

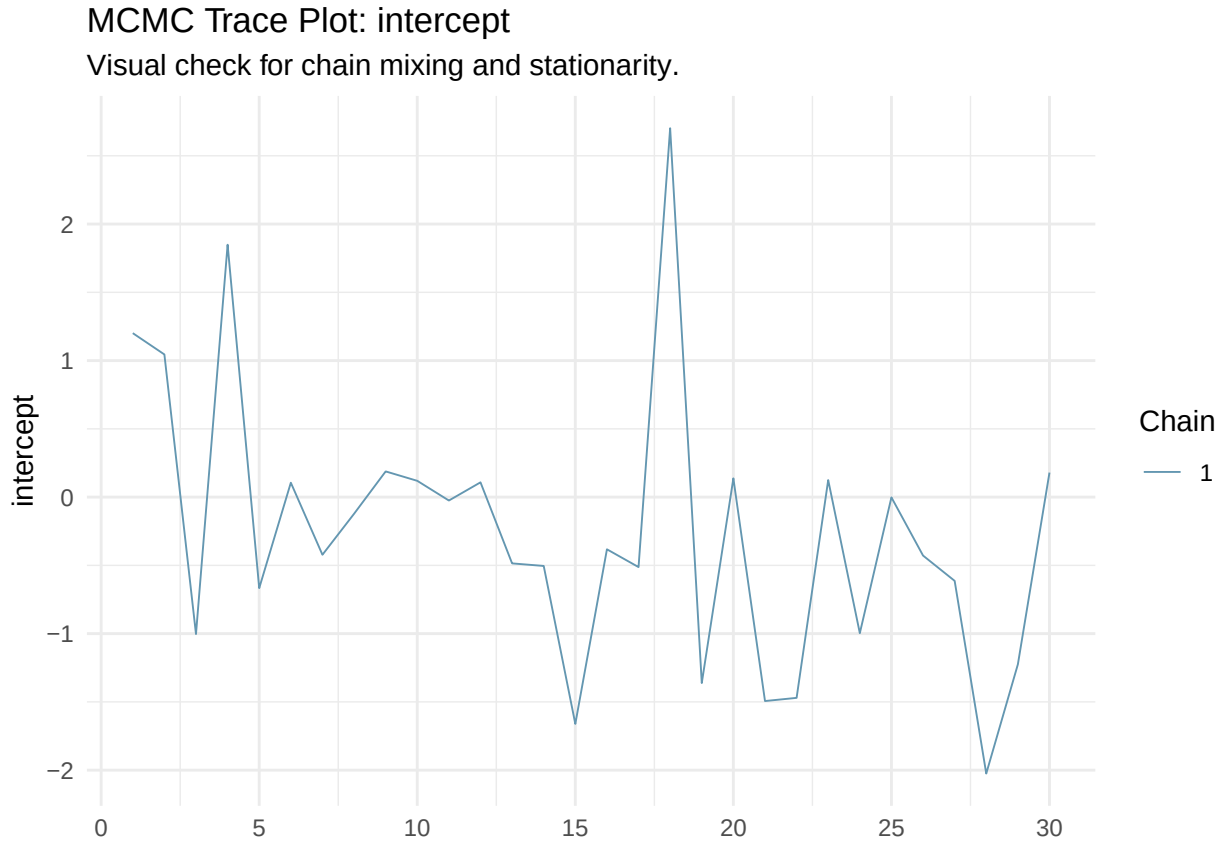
Statistical Summary

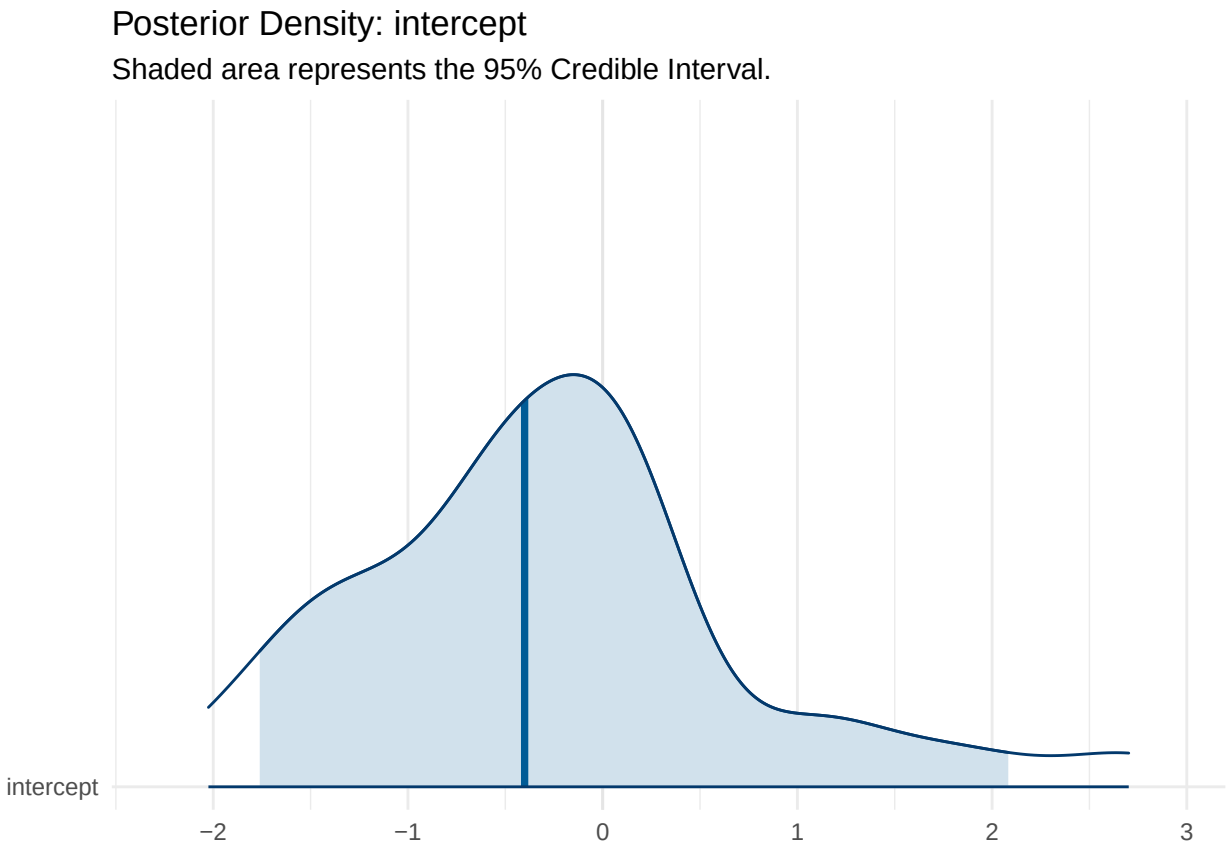
Table 1: Posterior Summary Statistics

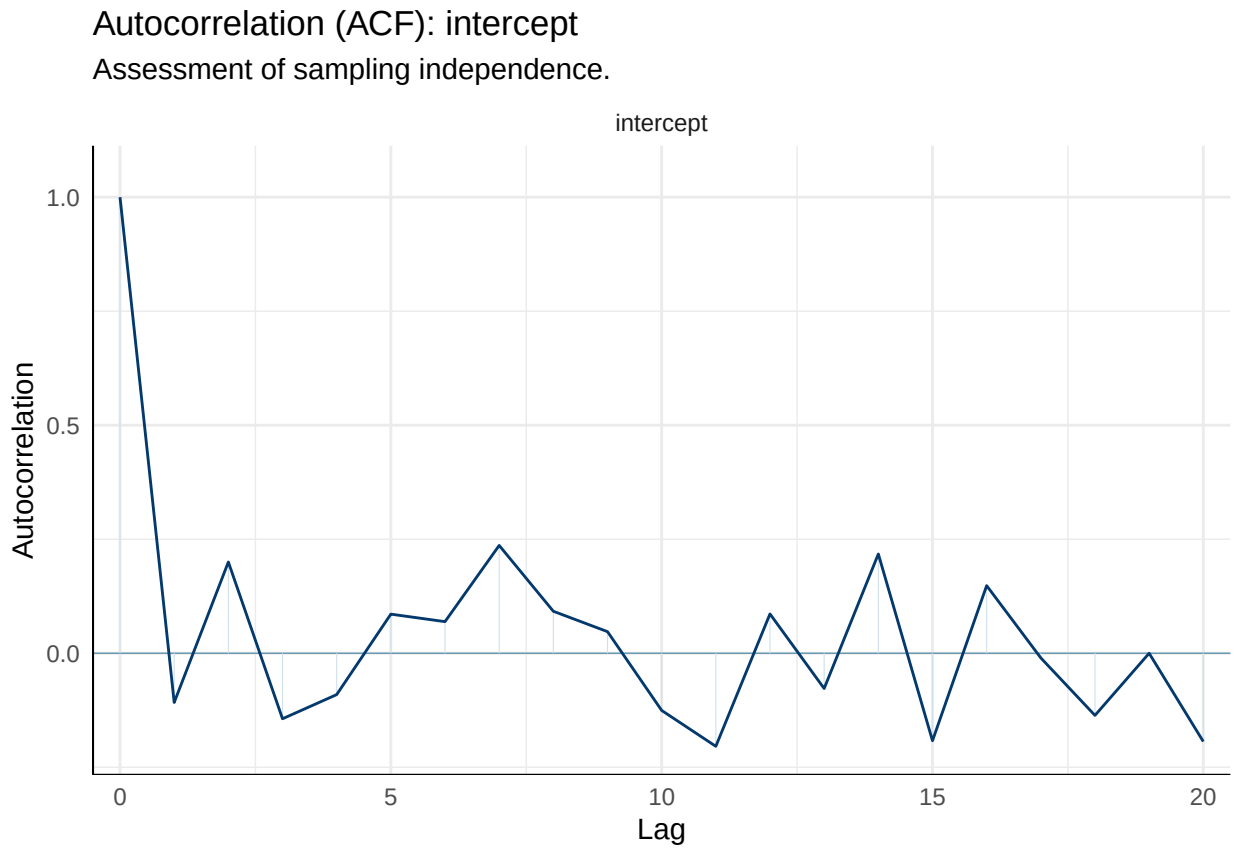
	Parameter	Mean	SD	2.5%	Median	97.5%	ESS
2.5%	intercept	-0.255	1.020	-1.761	-0.402	2.083	30
2.5%1	phi1	-0.079	0.814	-1.324	-0.195	1.330	30
2.5%2	sigma2	2.107	1.277	0.449	1.796	5.041	30

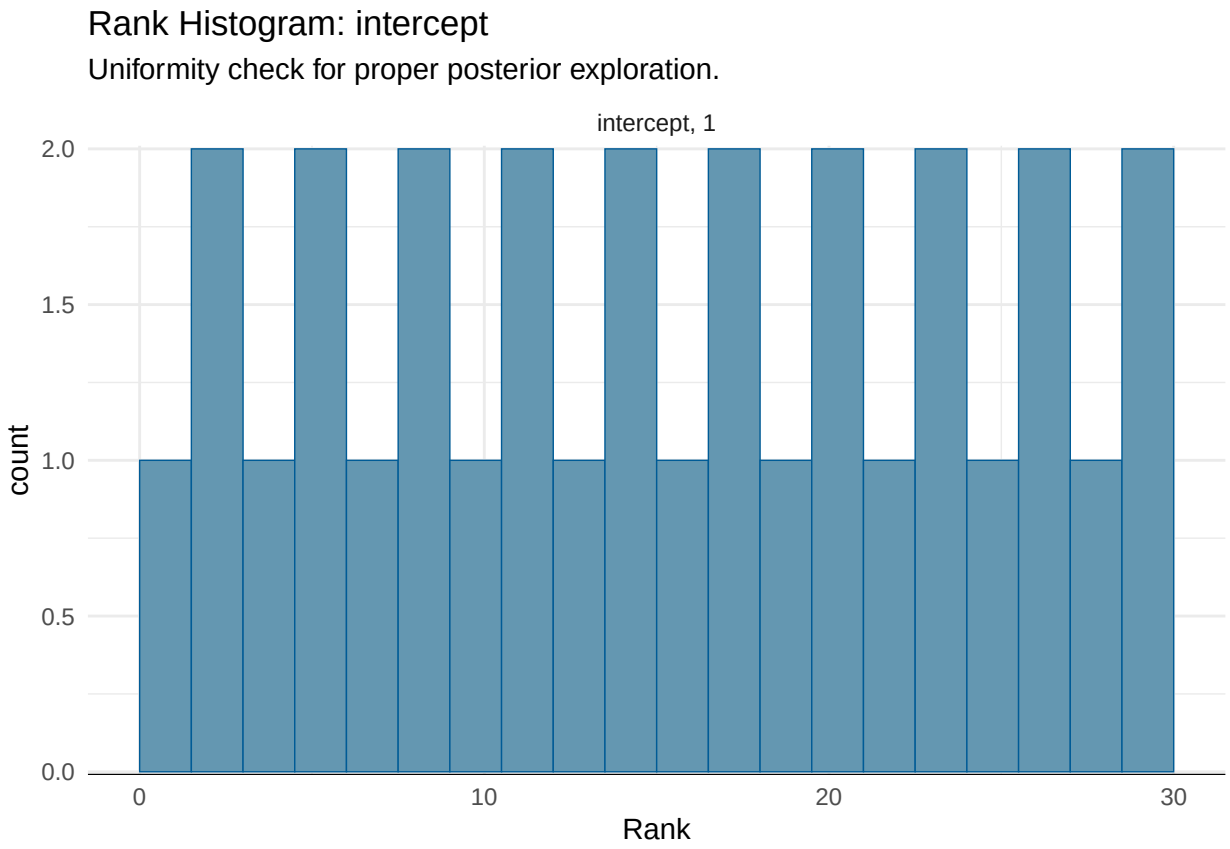
Parameter Diagnostics

Parameter: intercept

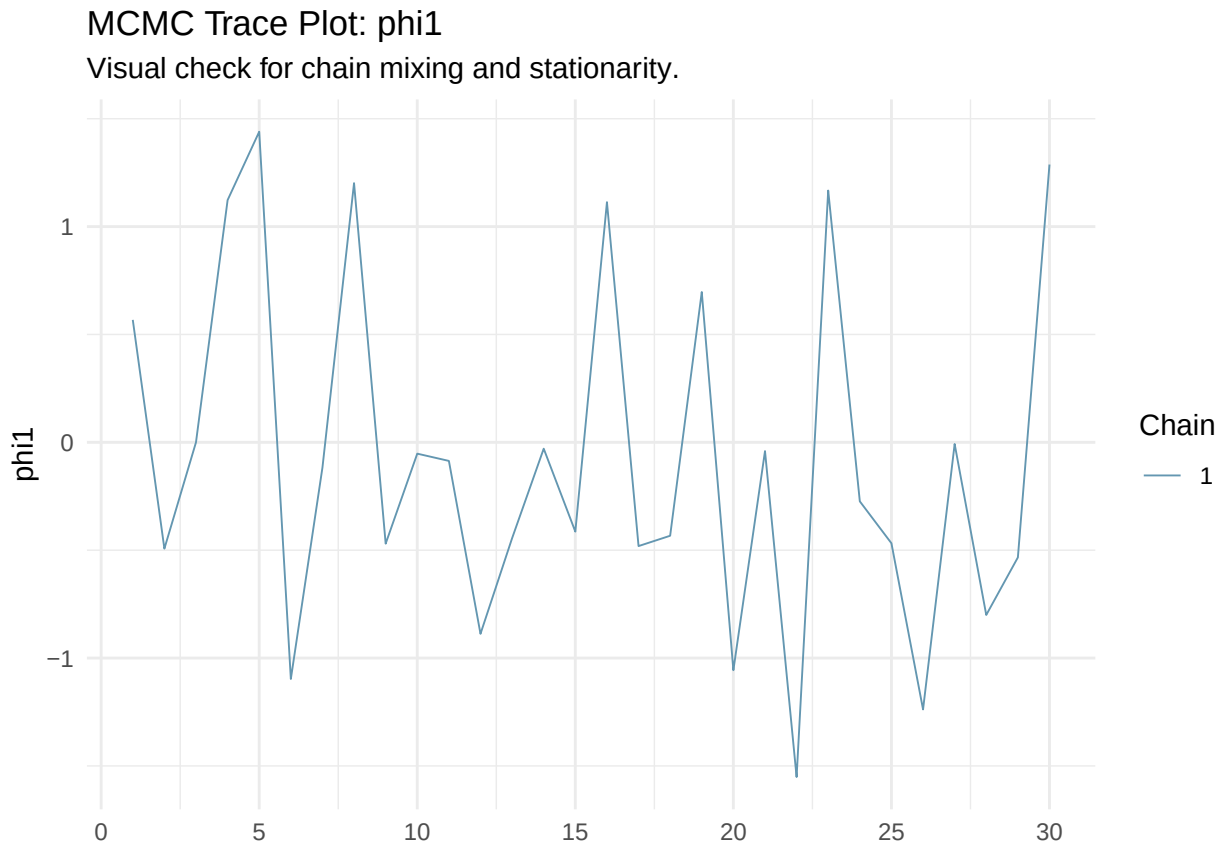


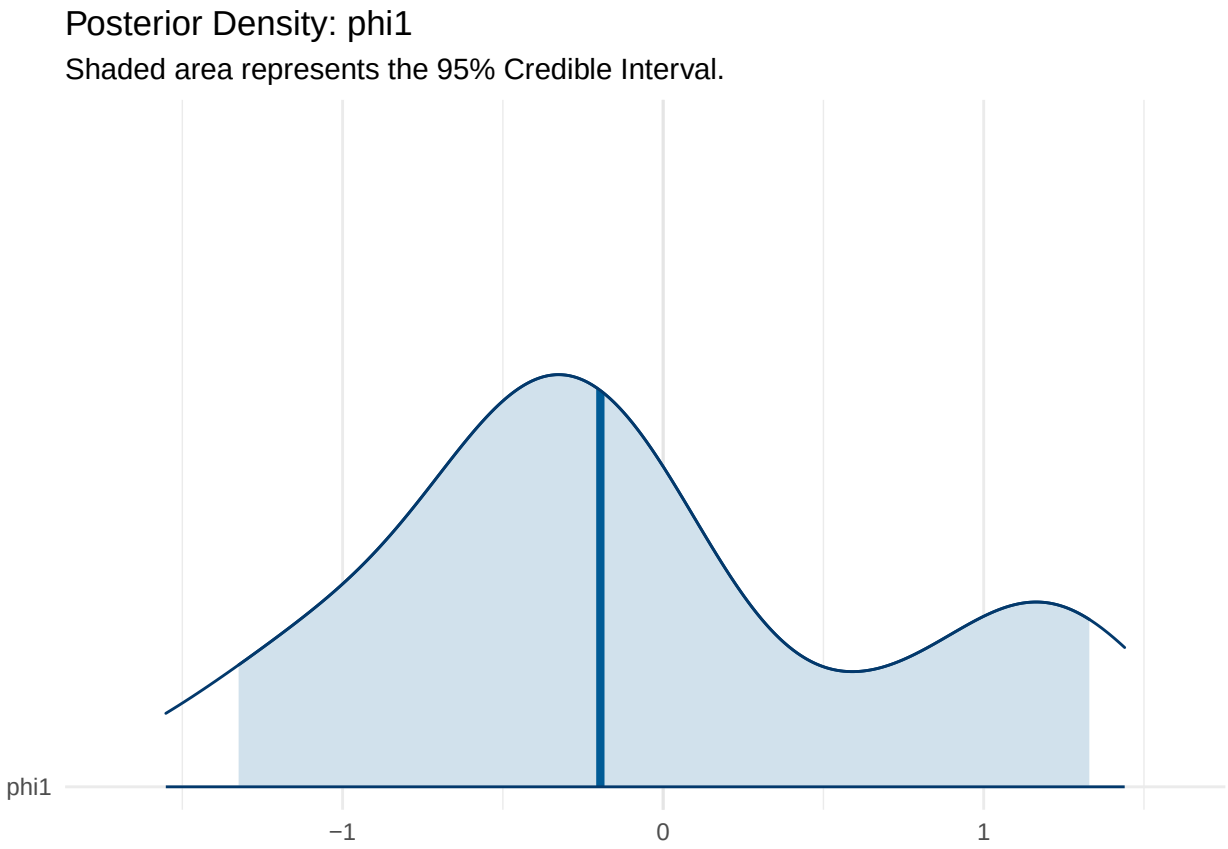

Posterior Density: intercept
Shaded area represents the 95% Credible Interval.
intercept
−2
−1
0
1
2
3

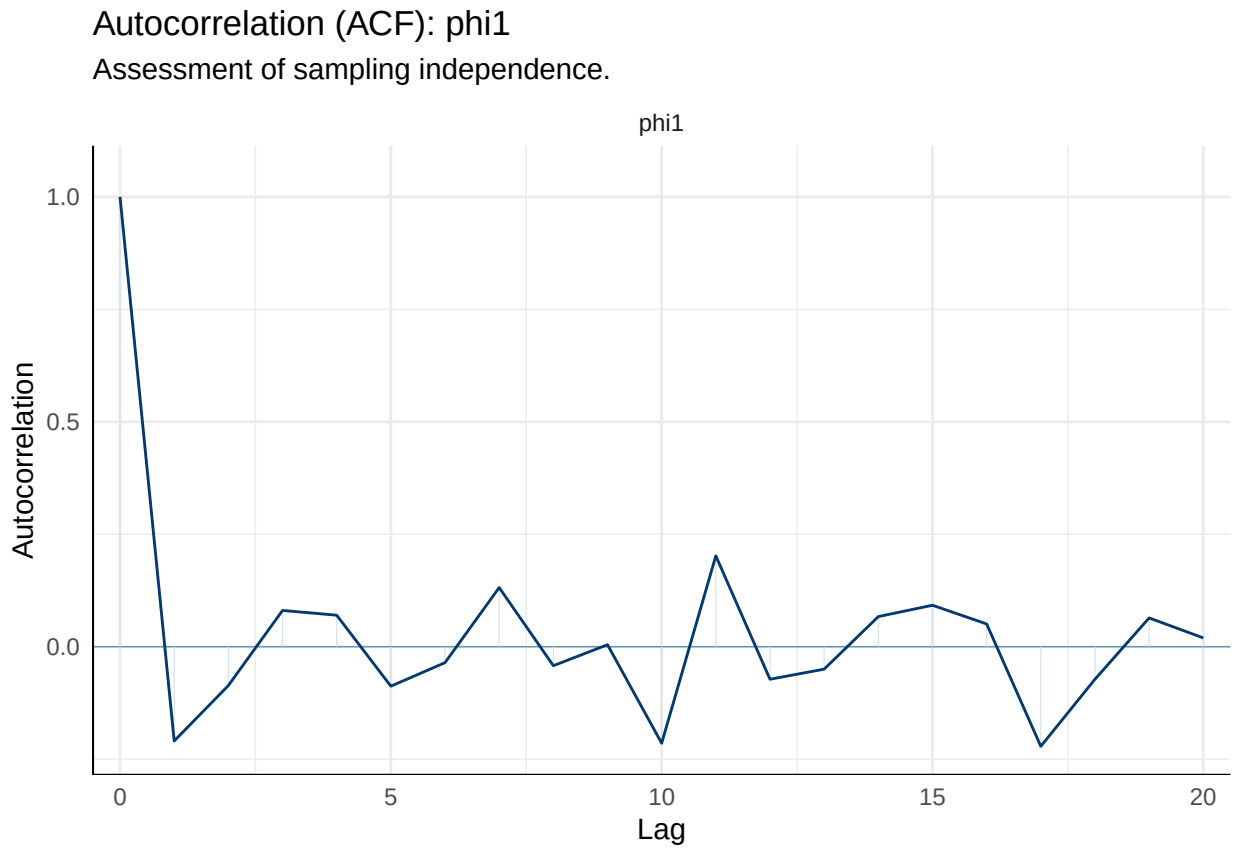

Autocorrelation (ACF): intercept
Assessment of sampling independence.
intercept
Autocorrelation
1.0
0.5
0.0
0
5
10
15
20
Lag

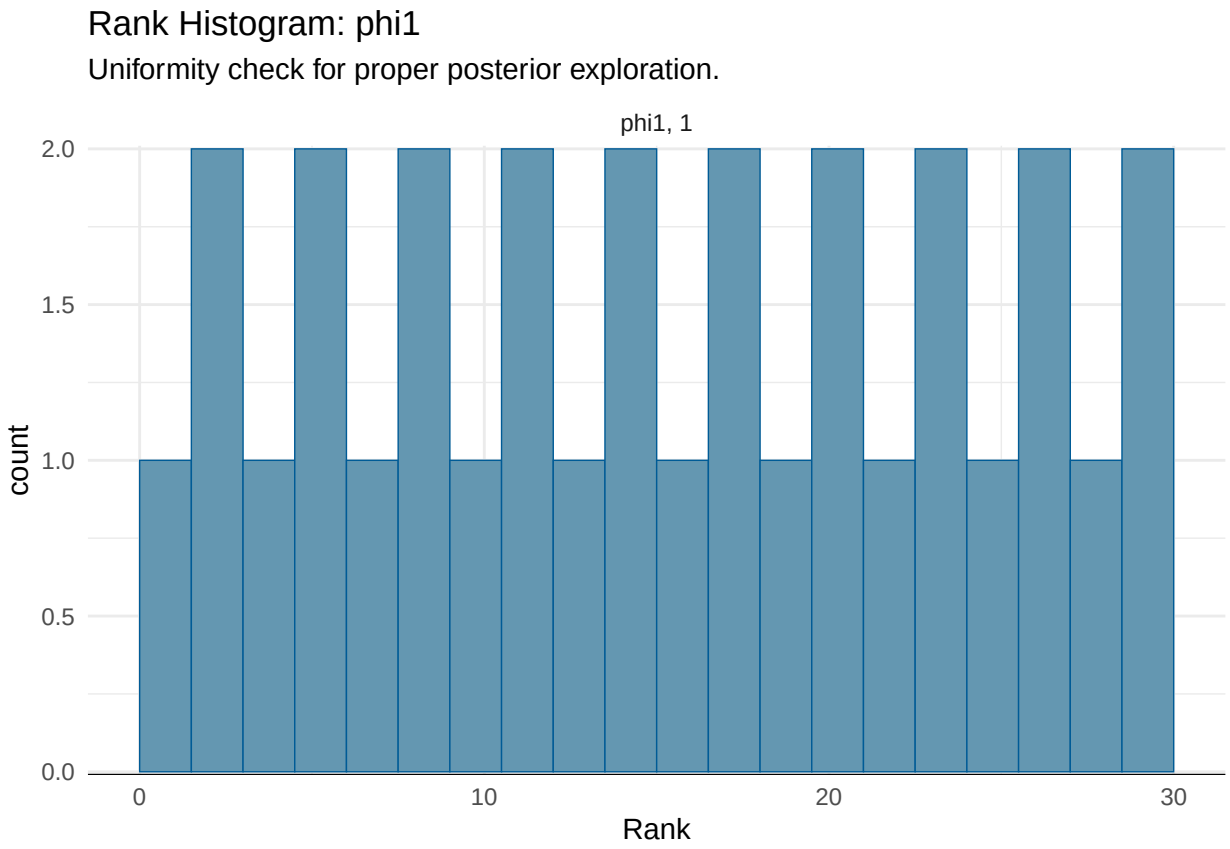

Rank Histogram: intercept
Uniformity check for proper posterior exploration.
intercept, 1
count
2.0
1.5
1.0
0.5
0.0
0
10
20
30
Rank

Parameter: phi1

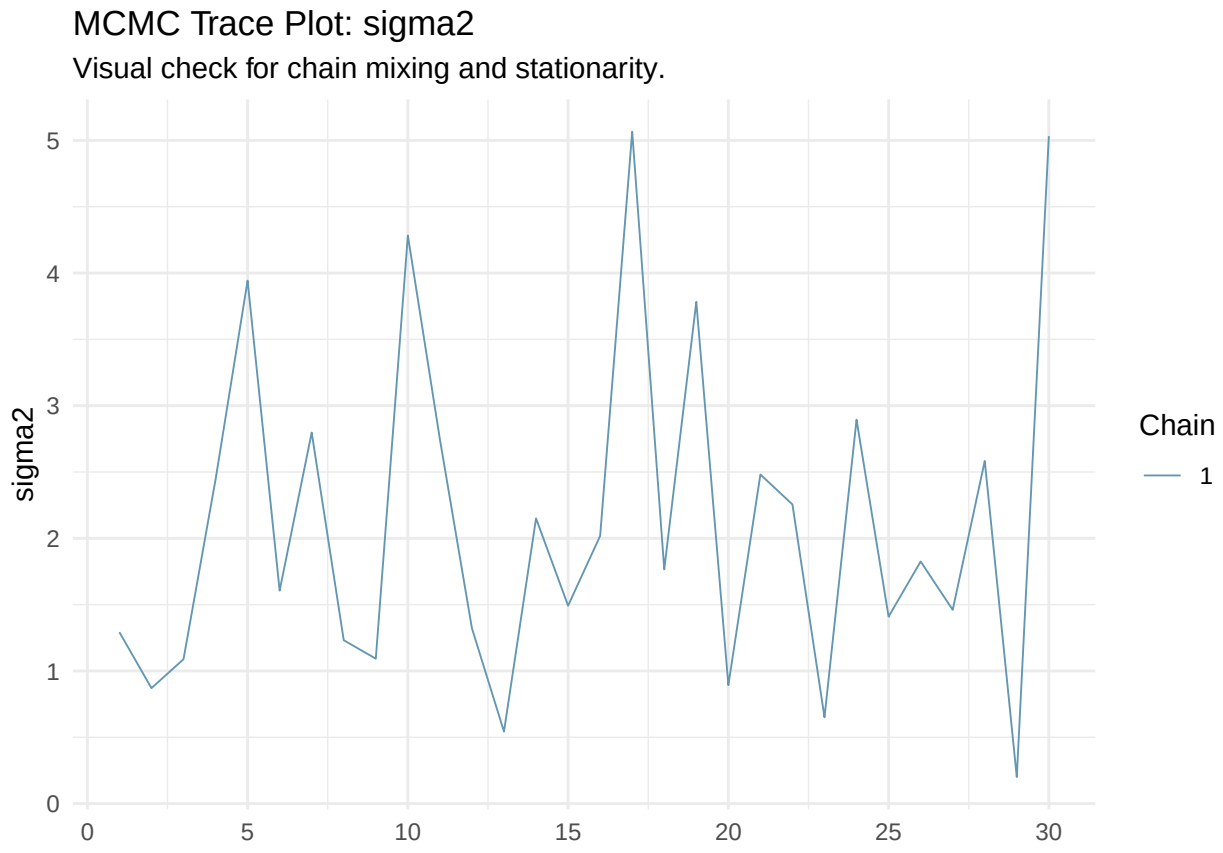


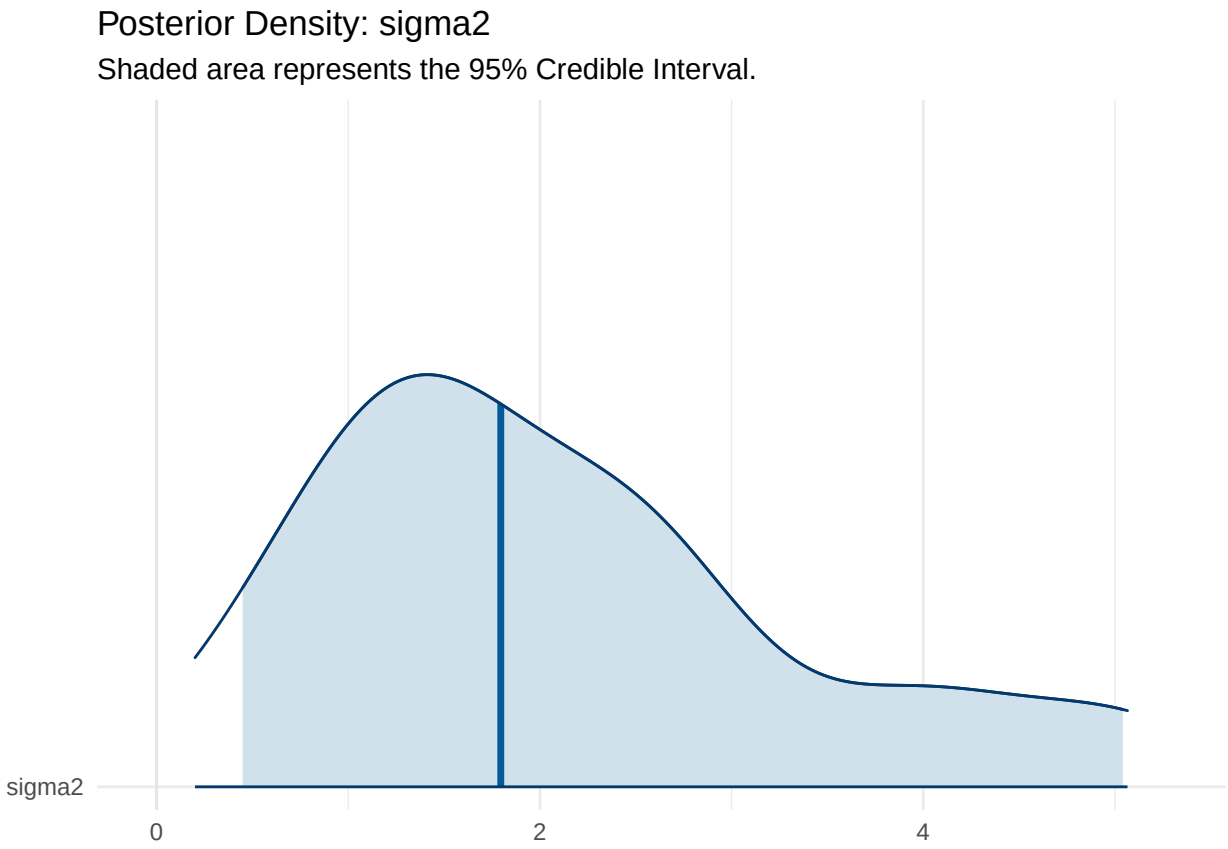

Posterior Density: phi1
Shaded area represents the 95% Credible Interval.
phi1
−1
0
1

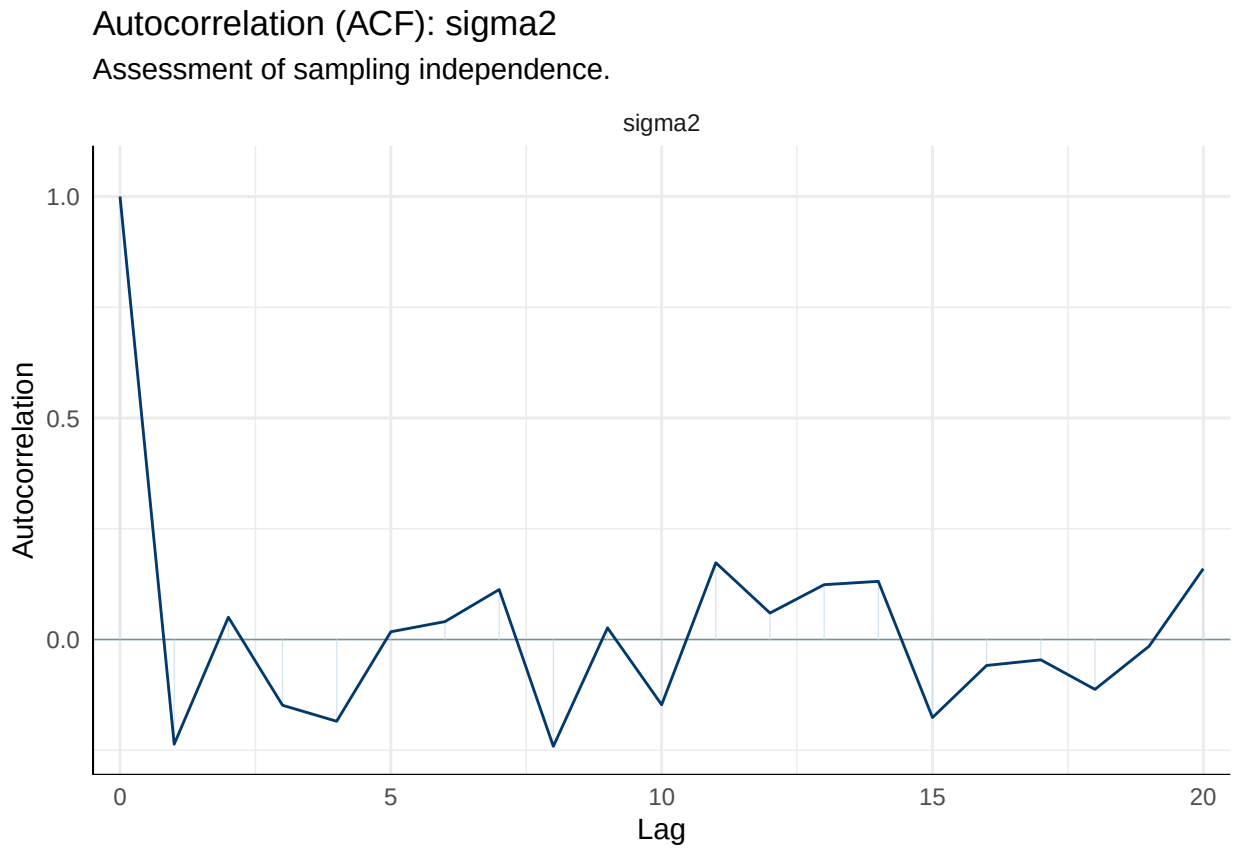

Autocorrelation (ACF): phi1
Assessment of sampling independence.
phi1
Autocorrelation
1.0
0.5
0.0
0
5
10
15
20
Lag

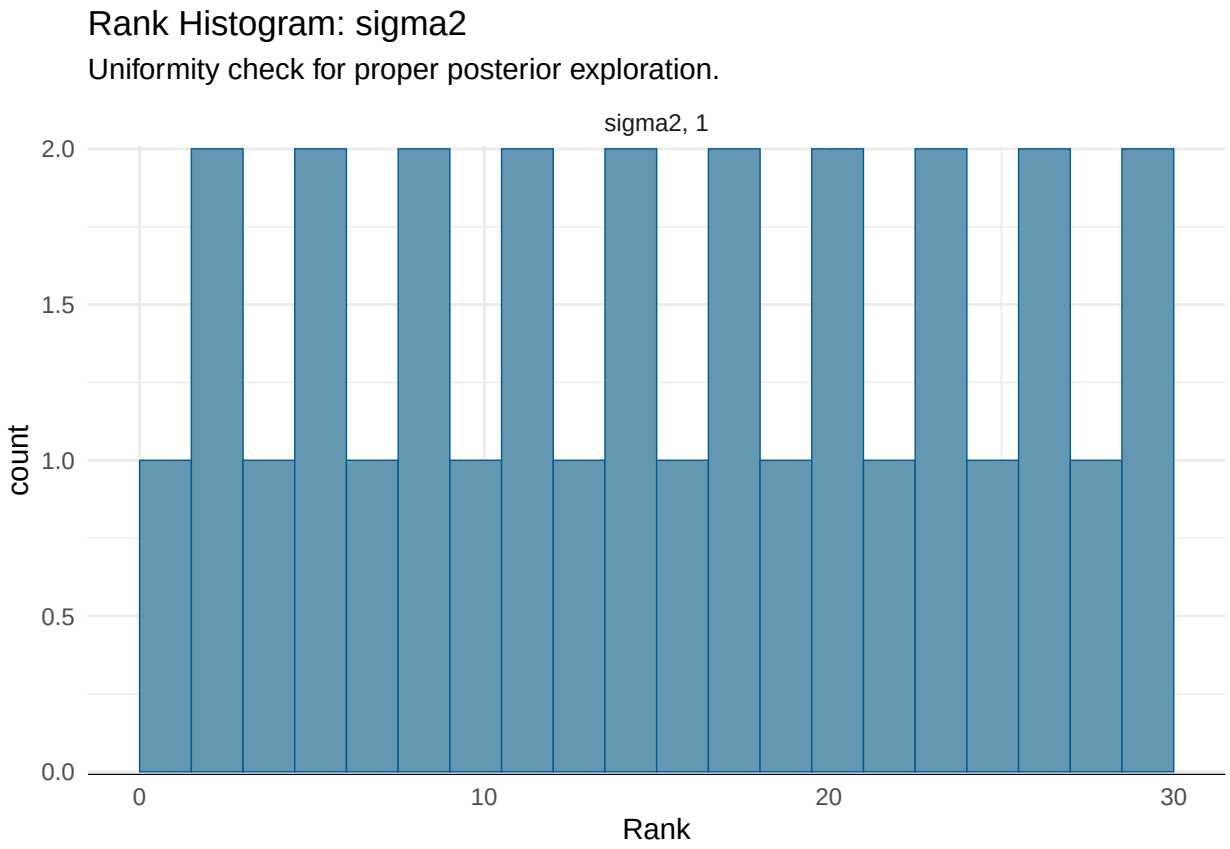

Rank Histogram: phi1
Uniformity check for proper posterior exploration.
phi1, 1
2.0
1.5
1.0
0.5
0.0
count
0
10
20
30
Rank

Parameter: sigma2




Posterior Density: sigma2
Shaded area represents the 95% Credible Interval.
sigma2
0
2
4


Autocorrelation (ACF): sigma2
Assessment of sampling independence.
sigma2
Autocorrelation
1.0
0.5
0.0
0
5
10
15
20
Lag


Rank Histogram: sigma2
Uniformity check for proper posterior exploration.
sigma2, 1
2.0
1.5
1.0
0.5
0.0
count
0
10
20
30
Rank

Glossary

- **ESS (Effective Sample Size):** Number of independent draws equivalent to the total samples.
- **95% HPD (Highest Posterior Density):** The narrowest interval containing 95% of the posterior mass.
- **Trace Plots:** Used to visually inspect convergence and mixing.
- **Rank Histograms:** Used to verify uniform exploration of the posterior logic.